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The Editors
General Relativity and Gravitation

Dear Editors,

Please consider my manuscript, “*Identifying a relaxing internal state in free-fall timing: finite-frequency boundaries and an application to PSR J0337+1715*”, as a Research article in *General Relativity and Gravitation*.

The paper asks what a free-fall timing experiment can identify when a body’s internal state relaxes on orbital timescales. Its contribution connects the effective description of compact bodies to the comparator and nuisance assumptions used in tests of gravity.

Proven. A nonzero single relaxation pole can be matched at K distinct positive carriers by a real derivative comparator of degree at most N exactly when $N \geq 2K - 1$. Positivity with a rate gap restricts this ambiguity. Two calibrated complex samples test for one observable relaxation time through a moment-variance equality, while finite precision and transient freedom limit the inference. These statements specialize classical interpolation and moment results, which the manuscript explicitly credits.

Counterexample candidate. A damped scalar-charge model gives reciprocal forces and a controlled one-pole reduction. **Proven.** Its leading physical drive has a phase closure that the dictionary in our earlier stored timing analysis violates. **Imported from prior work.** A conditional reanalysis reports the dependence on the nuisance span, calibrated uncertainty within specified covariance families, and the failure of that leading physical response at six evaluated lags. No relaxation is detected, and the excess in the archived carrier coefficients remains unattributed. **Conjectural.** Uncertain timing derivatives and incomplete physical matching prevent an empirical strong-equivalence-principle bound.

The paper fits the journal’s interest in gravitation and its experimental tests by making the assumptions behind dynamical identifiability explicit. Online Resource 1 provides the complete technical account, numerical audits and a worked white-dwarf calculation. The code, records and submission manifests are archived at <https://github.com/lpaiu-cs/self-mass-unobservability>, tag `grg-submission-2026-09-28`; the availability statement explains the omitted large arrays and the compact public replay.

AI agents were used substantively in coding, computation, drafting and internal review. Their use is disclosed in the manuscript, and I take responsibility for the work. The manuscript has not been published and is not under consideration elsewhere. I have no competing interests to declare, and this research received no external funding.

Sincerely,

Juneyoung Kim